

**Note : This is a three member team assignment – The team members can share their workload. Draw the diagrams neatly and if there are any derivations do it neatly. Theory portions in this assignment needs to be written in hand, photographed and converted to pdf and uploaded. Typewritten documents are NOT allowed for theory portions and will get very much reduced marks. [ below 50 percent]. Programming code and their explanation can be typewritten**

**Do NOT lend your documents for copying. If excessive similarity is found between documents, both the teams will be penalized.**

**Last date is 24-04-2025 23:59 hours. NO EXTENSION WILL BE GIVEN FOR ANY REASON. Late submissions will be rejected as COE office has advised us strictly that all internal evaluation will be completed before 26-04-2025.**

**ALL answers need to be combined and made as One submission. ONLY ONE SUBMISSION PER TEAM. Team members names and their registration numbers need to be specified very clearly in the front page. Multiple submissions will result in reduction of 1 mark for every member. Not putting the team members name in the front page and later claiming that some person is part of the team is STRICTLY NOT ALLOWED>**

PART-A Study about following and write in your own handwriting notes about the following.

1. JPEG compression technique and decompression technique – Block diagram and functionality of each block in a paragraph with associated formulae if any [Minimum 3 pages]
2. JPEG 2000 compression technique and decompression technique – Block diagram and functionality of each block in a paragraph with associated formulae if any.[Minimum 3 pages]
3. Comparison of JPEG and JPEG2000 – [ 1 page]
4. Scale Invariant Feature Transform [ SIFT] – All steps diagram and functionality [ Minimum 4 pages ]
5. Speeded Up Robust Features [ SURF] – All steps diagram and functionality[ Minimum 4 pages ]
6. Comparison of SIFT and SURF [ 1 page]

PART-B Numerical problems

7. HUFFMAN coding problem

**8.9** Consider the simple  $4 \times 8$ , 8-bit image:

|    |    |    |    |     |     |     |     |
|----|----|----|----|-----|-----|-----|-----|
| 21 | 21 | 21 | 95 | 169 | 243 | 243 | 243 |
| 21 | 21 | 21 | 95 | 169 | 243 | 243 | 243 |
| 21 | 21 | 21 | 95 | 169 | 243 | 243 | 243 |
| 21 | 21 | 21 | 95 | 169 | 243 | 243 | 243 |

- (a) Compute the entropy of the image.
- (b) Compress the image using Huffman coding.
- (c) Compute the compression achieved and the effectiveness of the Huffman coding.

8. ARITHMETIC ENCODING AND DECODING PROBLEM

**8.17** Given a four-symbol source  $\{a, b, c, d\}$  with source probabilities  $\{0.1, 0.4, 0.3, 0.2\}$ , arithmetically encode the sequence *bbadc*.

**8.18\*** The arithmetic decoding process is the reverse of the encoding procedure. Decode the message 0.23355 given the coding model

| Symbol | Probability |
|--------|-------------|
| a      | 0.2         |
| e      | 0.3         |
| i      | 0.1         |
| o      | 0.2         |
| u      | 0.1         |
| !      | 0.1         |

9. OTSU algorithm for segmentation problem:

In the table for Problem 7 above

diagram

[i] replace All 21s with ZEROes

[ii] replace All 95s with 1s

[iii] replace All 169s with 2s

[iv] replace All 243s with 3s

Use 2 bit digitization for intensity values. Use OTSU algorithm to

decide the best threshold. Find the goodness of that threshold.